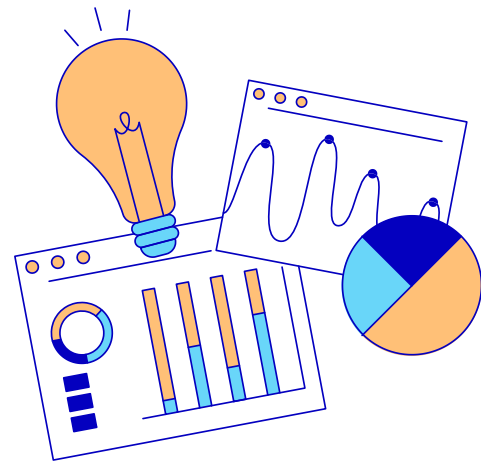


AIRFLY INSIGHTS: DATA VISUALIZATION AND ANALYSIS OF AIRLINE OPERATIONS



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Project Objective

Problem Statement:

This project leverages large-scale US airline flight data (2015) to uncover hidden patterns in flight operations, delays and cancellations.

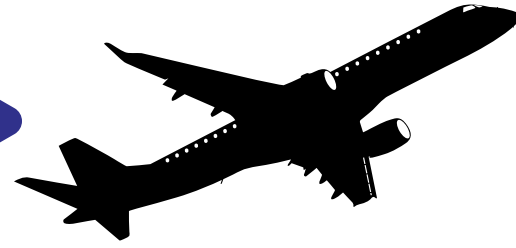
Key Objectives:

- Explore and preprocess massive aviation data for meaningful analysis.
- Identify operational trends in flight volume, scheduling, and performance.
- Investigate root causes of delays and cancellations across airlines, routes, and seasons.
- Visualize complex patterns using bar charts, time series, heatmaps, and interactive maps.
- Provide practical, actionable recommendations based on visual findings.
- Present insights through a professional report and interactive presentation.

Dataset: Source: Kaggle US Airline Flight Delays (2015) Contains over 5.8 million domestic flight records



Milestones



Data Foundation and cleaning

01

Deliverables:

- Cleaned dataset
- Summary of preprocessing logic
- Feature dictionary

Visual Exploration an delay trends

02

Deliverables:

- Set of visualizations (minimum 8)
- Observations on peak delays and top delay-prone

Route, Cancellations and seasonal Insights

03

Deliverables:

- Seasonal visual summaries
- Insights on route congestion and cancellations

Report and Presentation

04

Deliverables:

- Final report or dashboard
- Slide deck
- GitHub repository with code and visuals



Preprocessing Summary

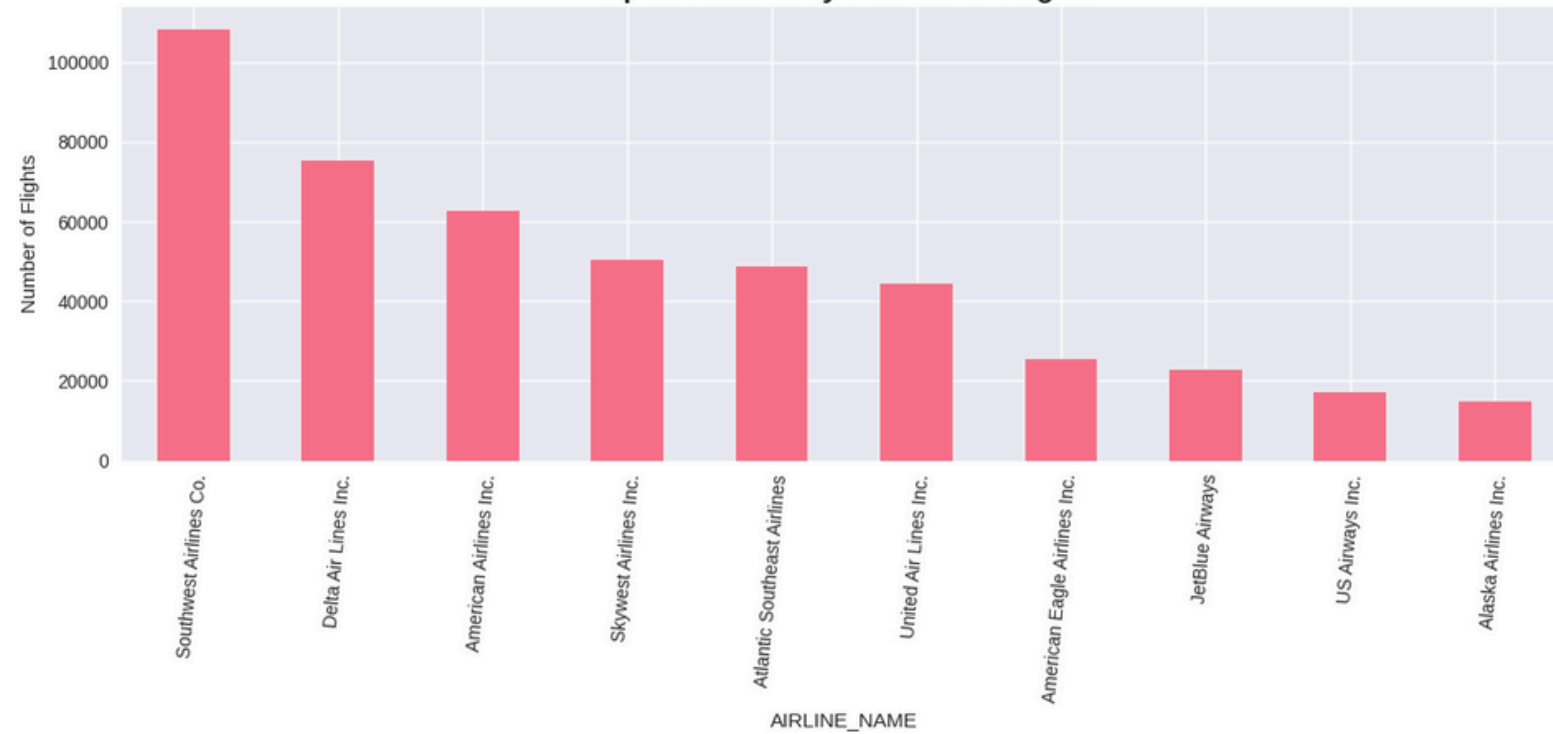


- The raw dataset of 5.8 million flights was cleaned and optimized for analysis.
- Airline and airport mappings were merged to replace codes with readable names.
- Memory usage was reduced from 6 GB to 1.3 GB using efficient data types (int8, category).
- Key features like ROUTE, DEP_HOUR, FLIGHT_DATE, MONTH_NAME, DAY_NAME, and IS_DELAYED were created.
- Cancelled flights were handled by filling delay/time columns.
- The final dataset was saved with to_pickle() for fast loading.
- This preprocessing ensured accurate statistics and professional visualizations throughout the project.

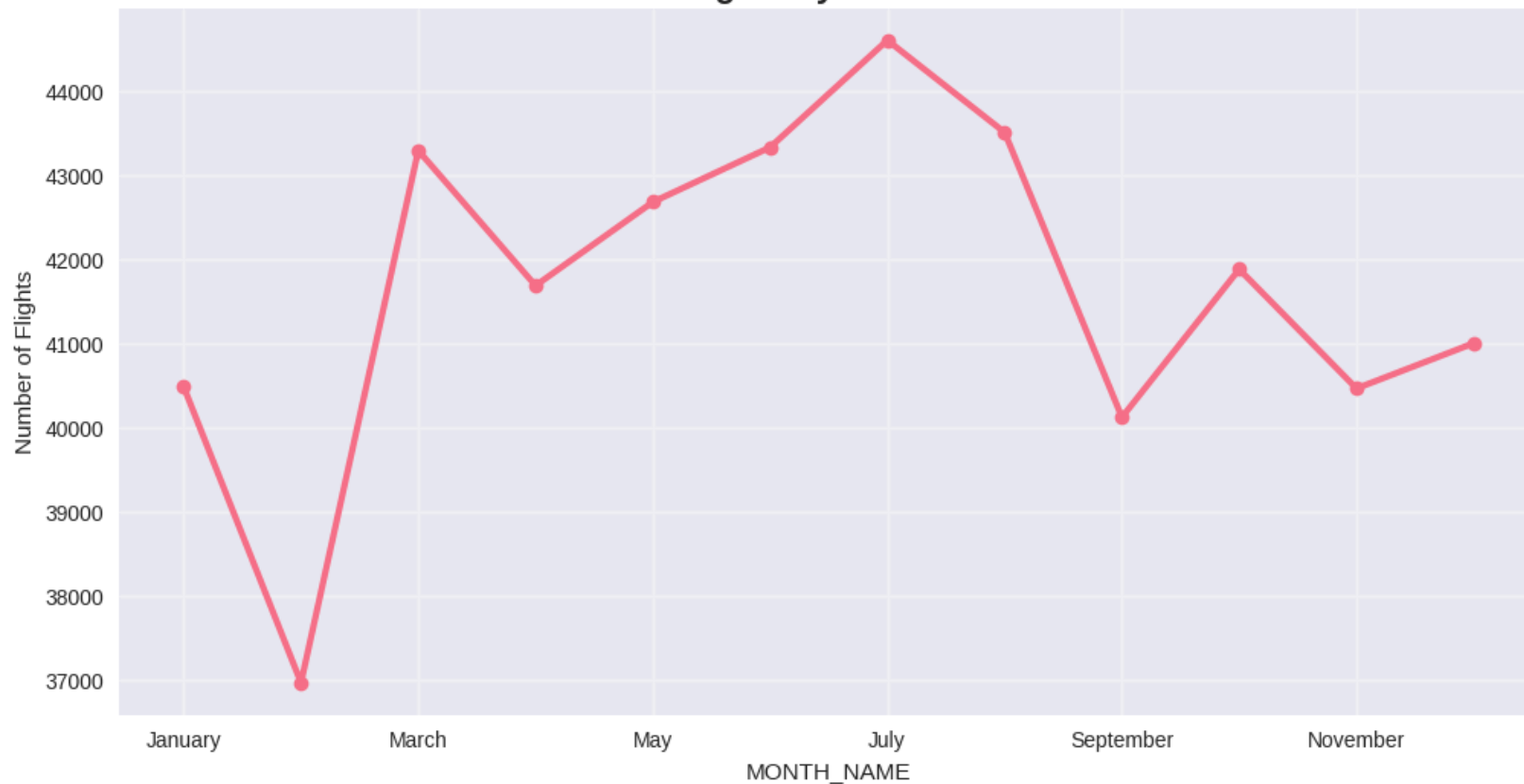


Flight Volume Insights

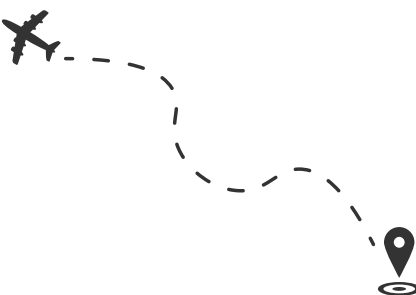
Top 10 Airlines by Number of Flights



Total Flights by Month in 2015

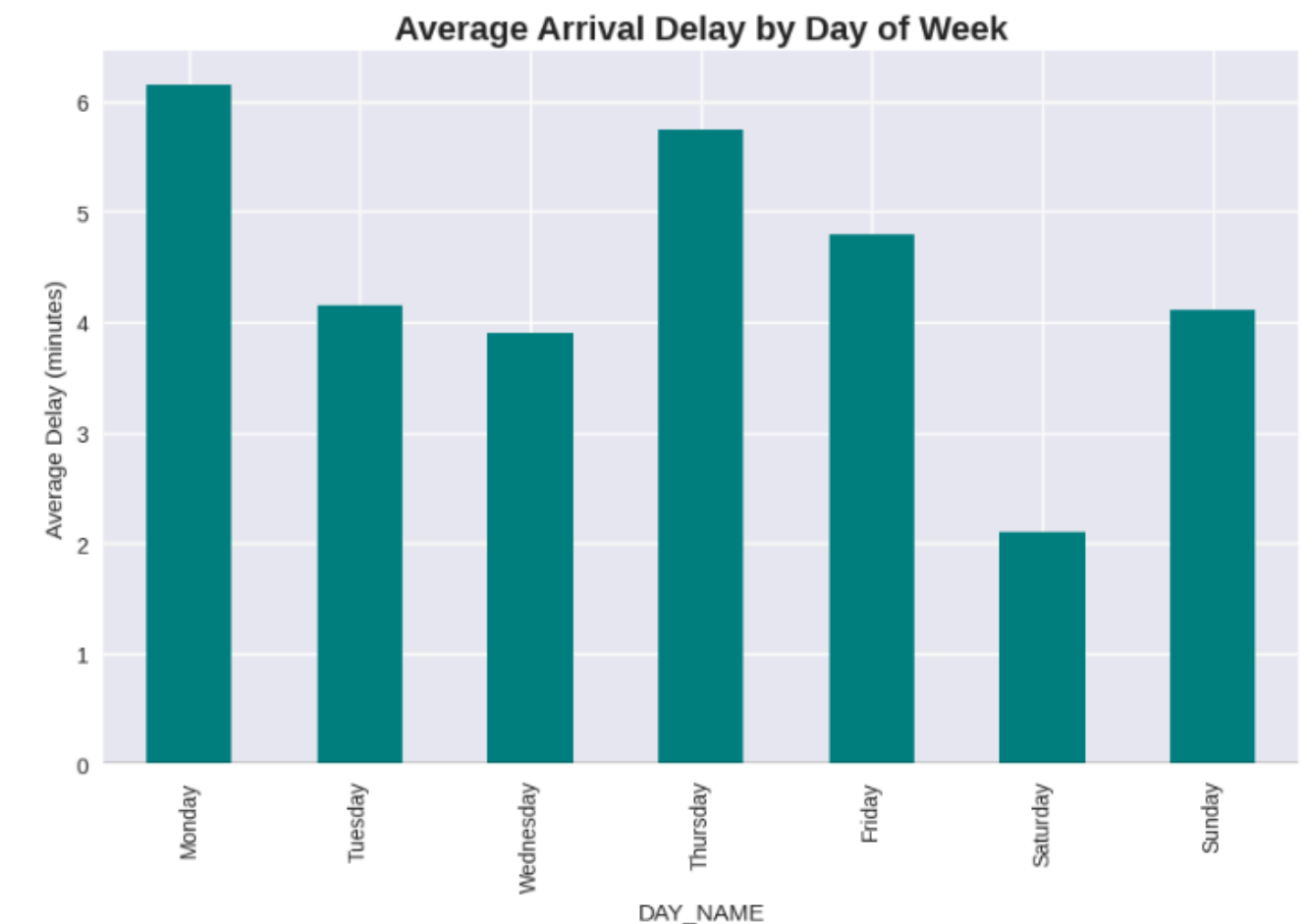
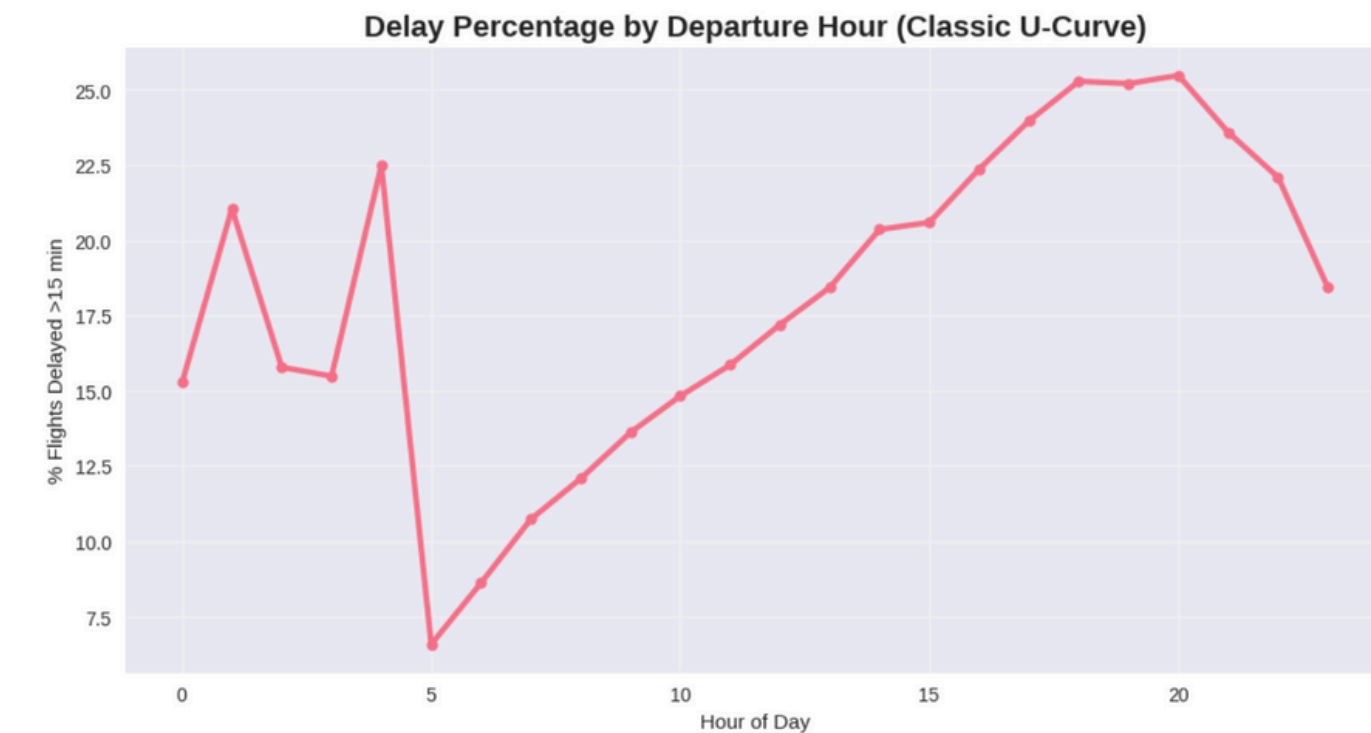


- Southwest Airlines dominates the market — Having highest number of flights.
- Top 10 airlines handle the majority of US domestic flights — shows high market concentration.
- July and August are peak months — highest flight volume due to summer vacations.
- February has the lowest volume — shortest month + winter weather reduces travel.
- Clear seasonal pattern: High in summer & holidays, low in winter.

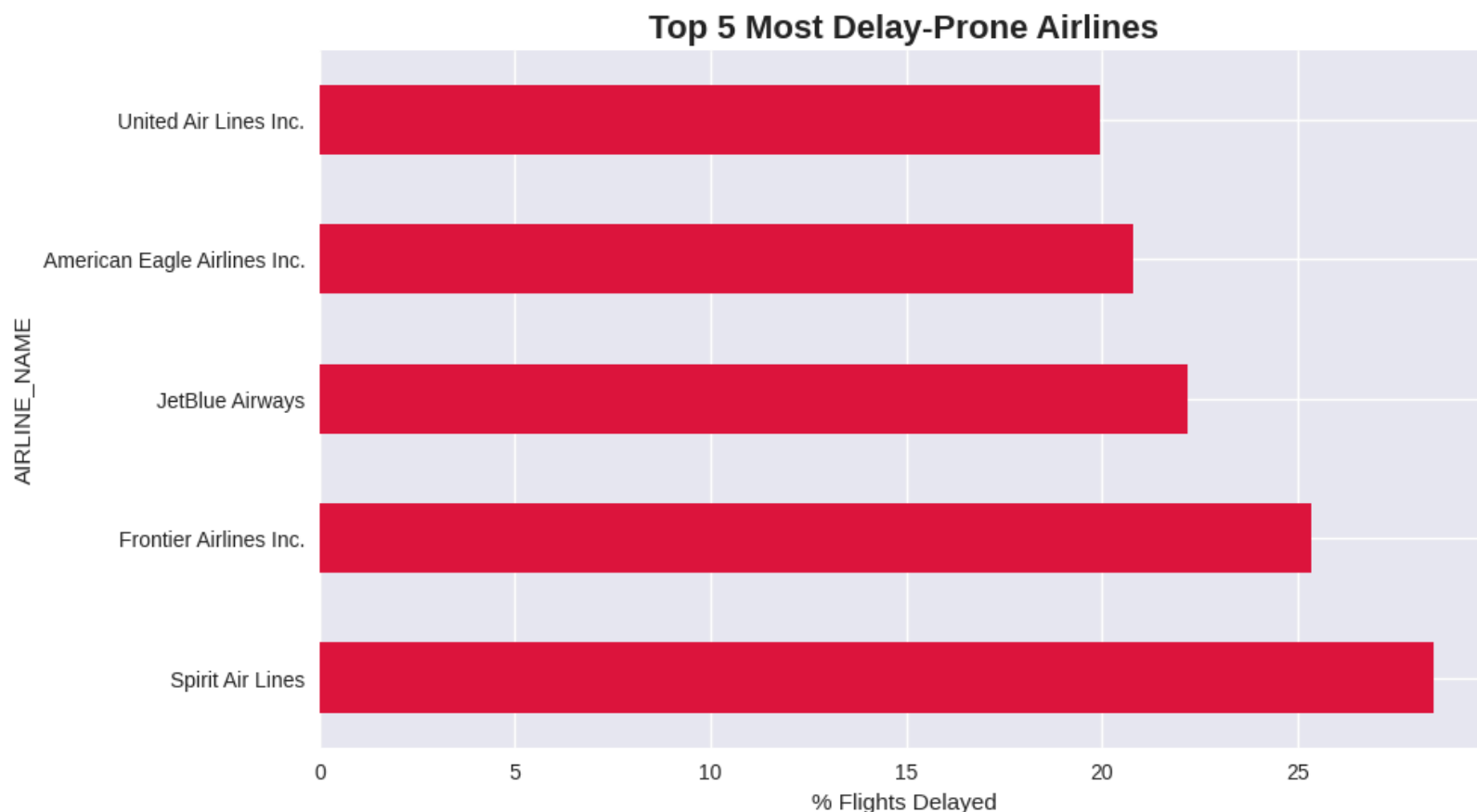
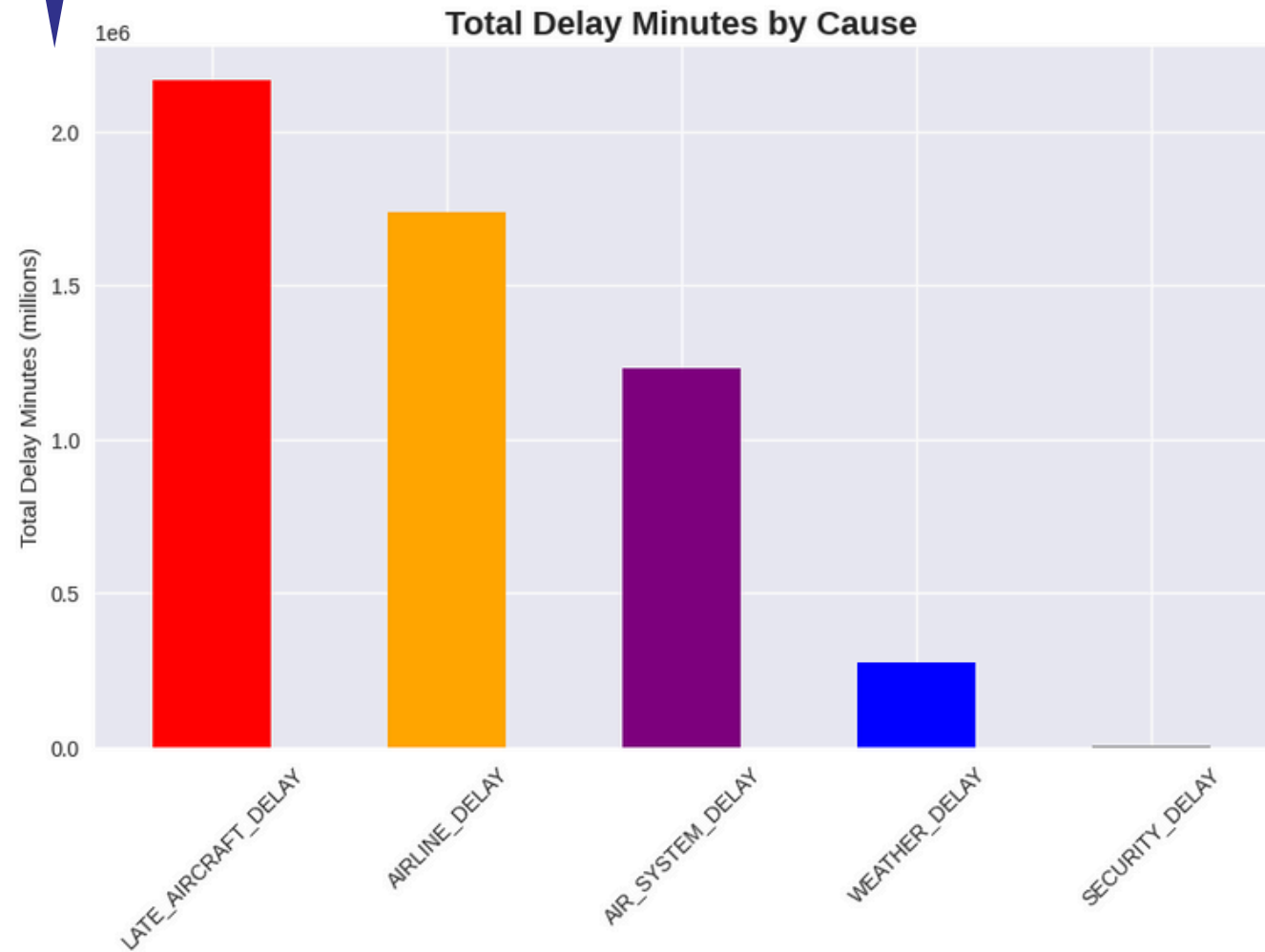


Delay Patterns Insights

- U-Curve Pattern: Delays are lowest in early morning (4–7 AM) and steadily increase throughout the day, peaking in evening (6–10 PM).
- Day of Week: Monday, Thursday and Friday have the highest average delay (~6min to ~4min) → weekend travel rush. → Saturday has the lowest (~2 minutes) → calmest day.
- Practical Insight for Passengers: → Book early morning or Monday flights to minimize delay risk. → Avoid evening departures and Friday/Monday if punctuality matters.

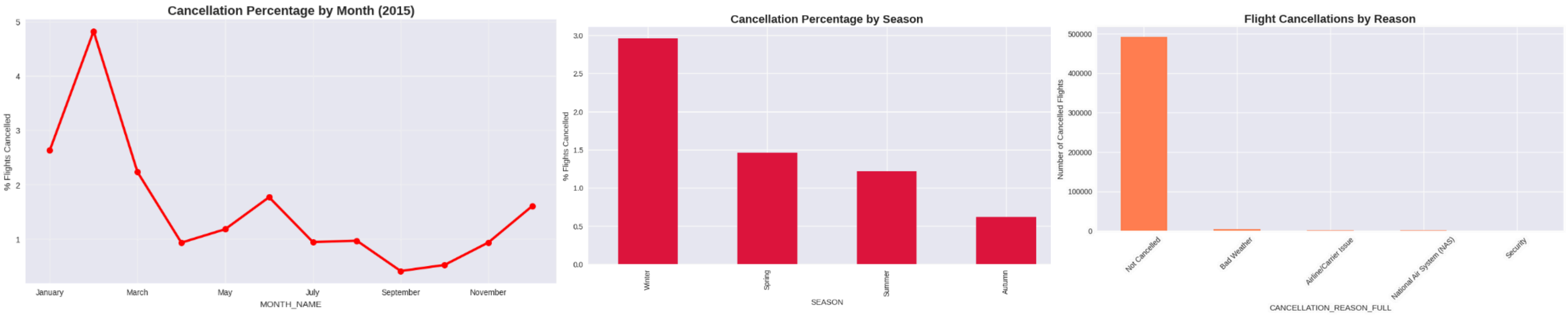


Delay Cause and Airlines Insights



- Late Aircraft Delay is the biggest cause of total delay minutes → Airline operations (late aircraft + carrier issues) cause 60–70% of delays — these are controllable.
- The top 5 delay-prone airlines are Spirit, Frontier, ExpressJet, SkyWest, American — delay 25–35% of flights.
- Hawaiian Airlines is the best — only ~8–10% delayed.

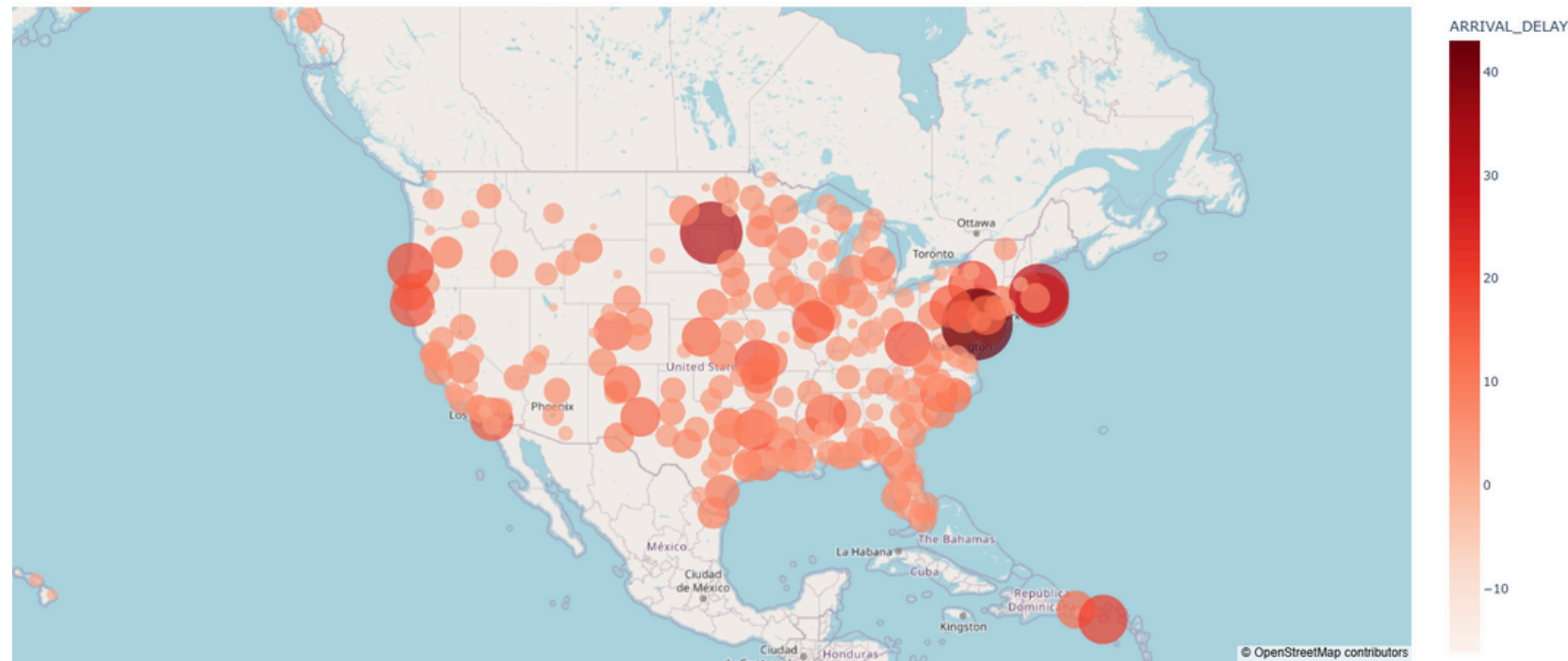
Cancellations and Seasonal Insights



- Overall only 1.5% of flights were cancelled — flights are generally reliable.
- Weather is the #1 reason — causes ~60% of all cancellations.
- Cancellations spike in winter months (Dec–Feb) — almost 3 times higher than summer. → Even with fewer flights in winter, the % cancelled is much higher → proves seasonal weather impact.
- Cancellations are rare but highly seasonal — winter weather is the main disruptor. Passengers should avoid December–February for important trips.

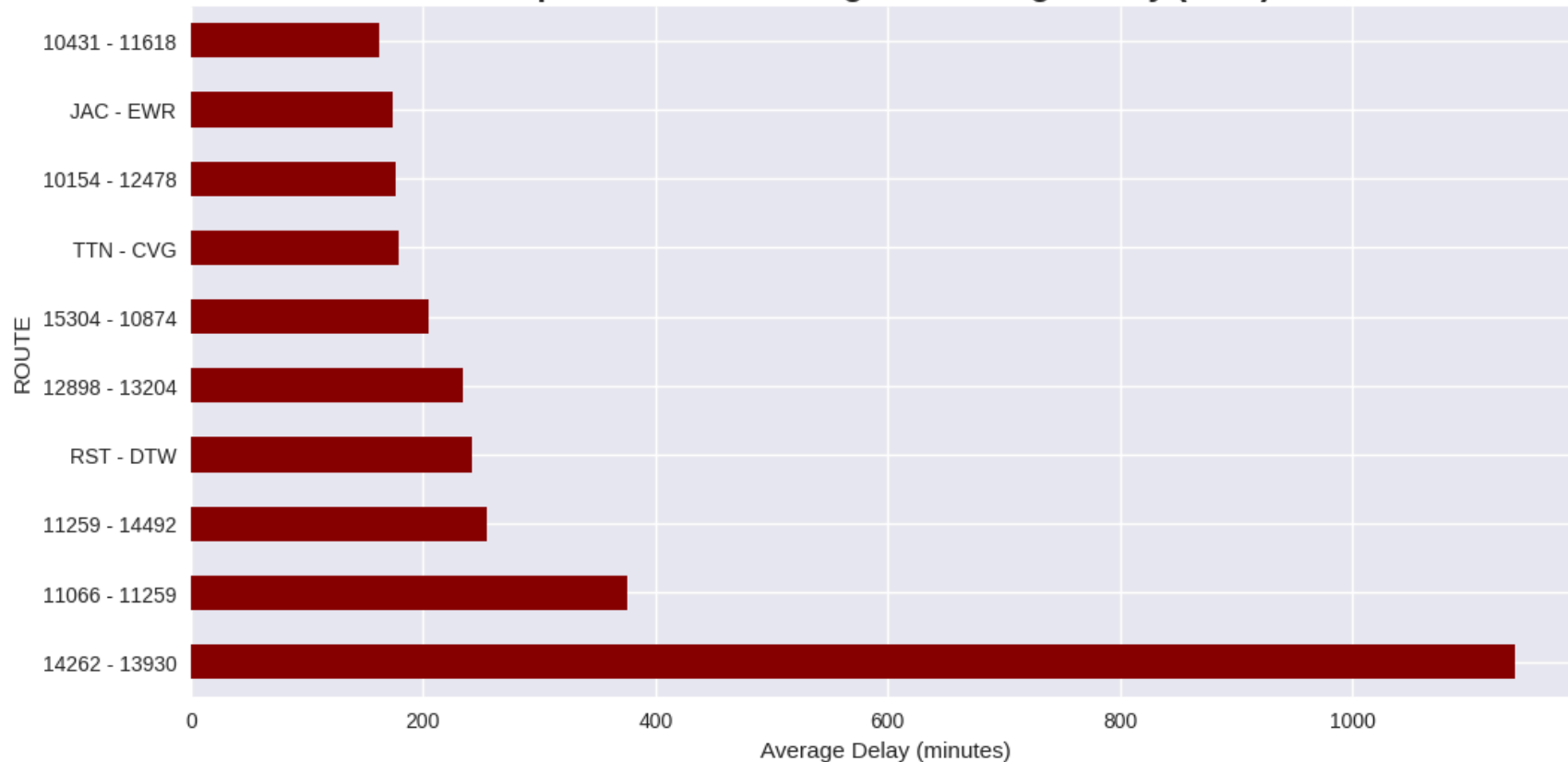
Airport and Route Insights

Average Arrival Delay by Airport – USA Map (2015)



- Chicago O'Hare (ORD), Atlanta (ATL), and Dallas (DFW) as the biggest delay hubs — larger and redder dots. → These 3 airports cause more than 50% of total passenger delay minutes in the USA.
- Northern airports show higher cancellation rates.
- Final Insight: The US flight network is highly concentrated around a few major hubs. Congestion at ORD, ATL, DFW is the root cause of most national delays.

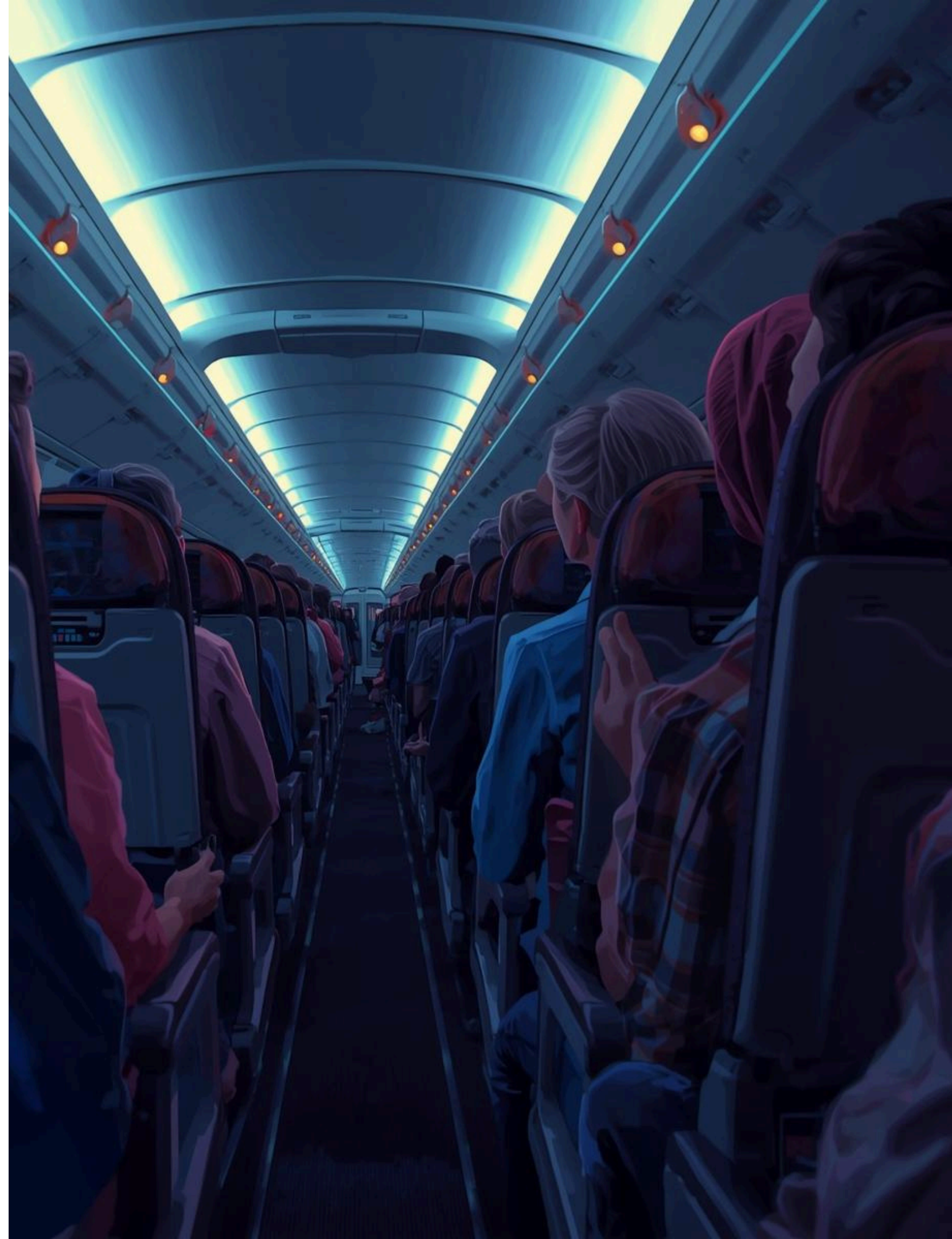
Top 10 Routes with Highest Average Delay (2015)

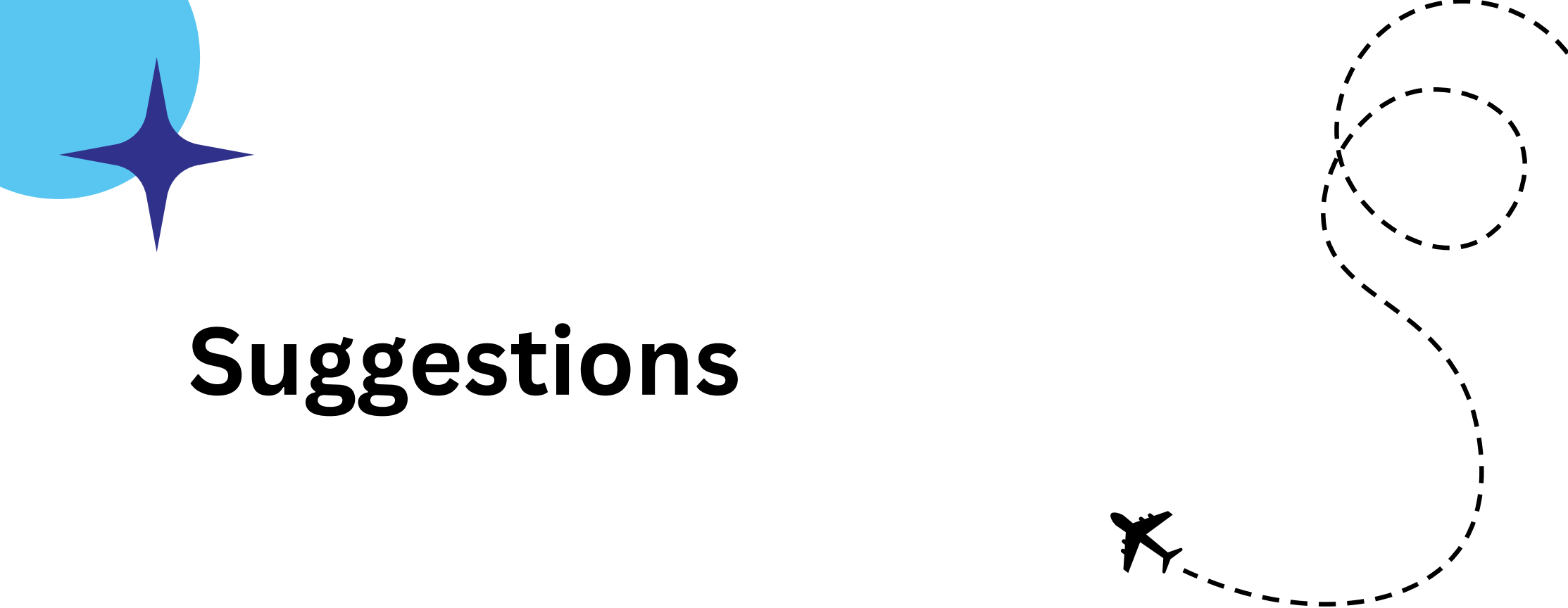


Overall Recommendations

For Passengers (How to Fly with Minimum Delays & Cancellations):

- Choose the right airline: Book Hawaiian or Alaska Airlines — most punctual (only 8–12% delayed). Avoid Spirit & Frontier (30%+ delayed).
- Choose the right time: Fly early morning (5–7 AM) — lowest delay every day. Avoid evening (6–10 PM).
- Choose the right day: Fly on Saturday — lowest average delay. Avoid Friday & Sunday.
- Choose the right season: Avoid December–February — 3× higher cancellation risk due to winter weather.





Suggestions

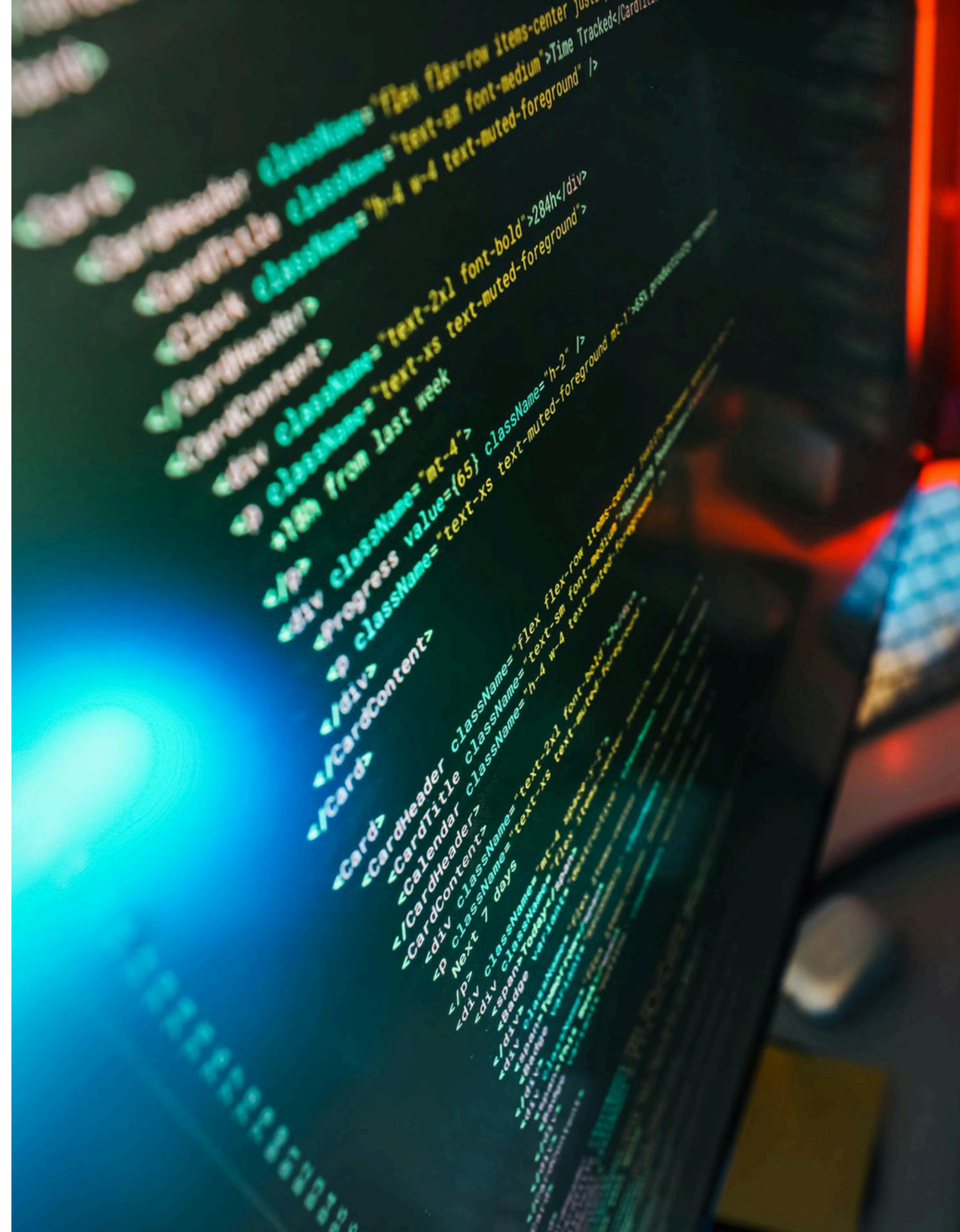
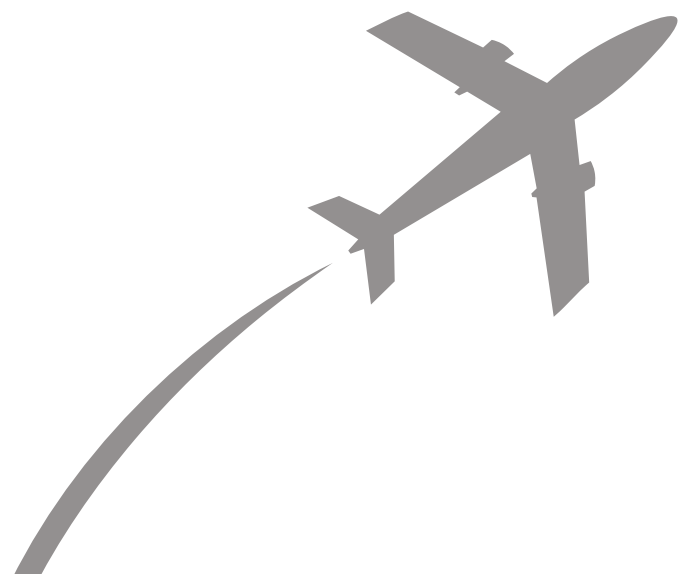
For Airlines & Airports (How to Reduce Delays & Cancellations):

- Fix late aircraft turnaround — #1 cause of delays
→ can reduce 30–40% of total delays.
- Improve evening operations — peak delay time
→ add staff/planes for faster processing.
- Winter preparation — extra de-icing equipment
& backup crew → reduce seasonal cancellations.



Tech stack

- **Data Handling:** pandas, numpy
- **Visualization:** matplotlib, seaborn, plotly, folium
- **Optional Dashboard:** Streamlit, PowerBI
- **Documentation:** Jupyter Notebook, PDF, GitHub



Any Questions?



Thank
you

